Serialize and Deserialize Binary Tree (hard)

Problem Statement

`Serialization` involves converting a binary tree into a string format such that it can be stored in the file or memory, preserving its structure and data.

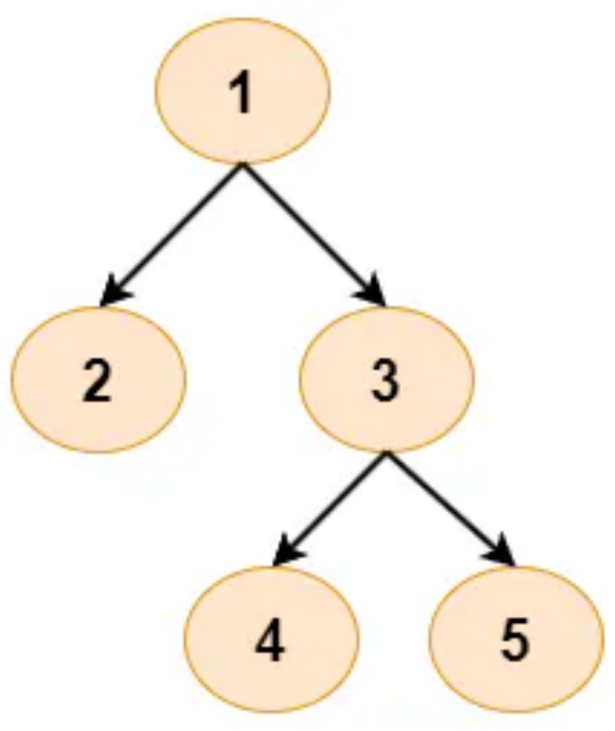
`Deserialization`, on the other hand, is about reconstructing the binary tree from this string representation.

Design an algorithm to `serialize` and `deserialize` the binary tree. There is no restriction on designing the algorithm for serialization and deserialization of the binary tree.

Examples

Example 1:

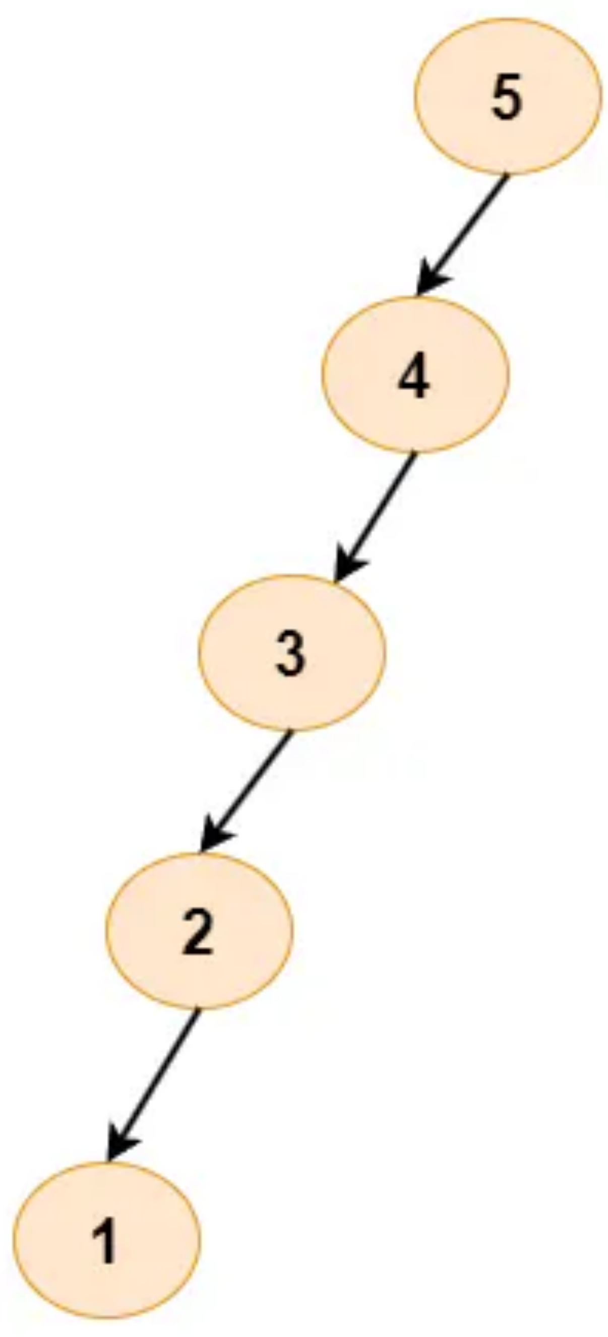
- **Input**: `[1,2,3,null,null,4,5]`



- **Expected Output**: `[1,2,3,null,null,4,5]`
- **Justification**: The tree starts with the root node 1, followed by its left child 2 and right child 3. Since 2 and 3 have no left children, they are followed by `null`. Nodes 4 and 5 are the left and right children of 3.

Example 2:

- **Input**: `[5,4,null,3,null,2,null,1]`



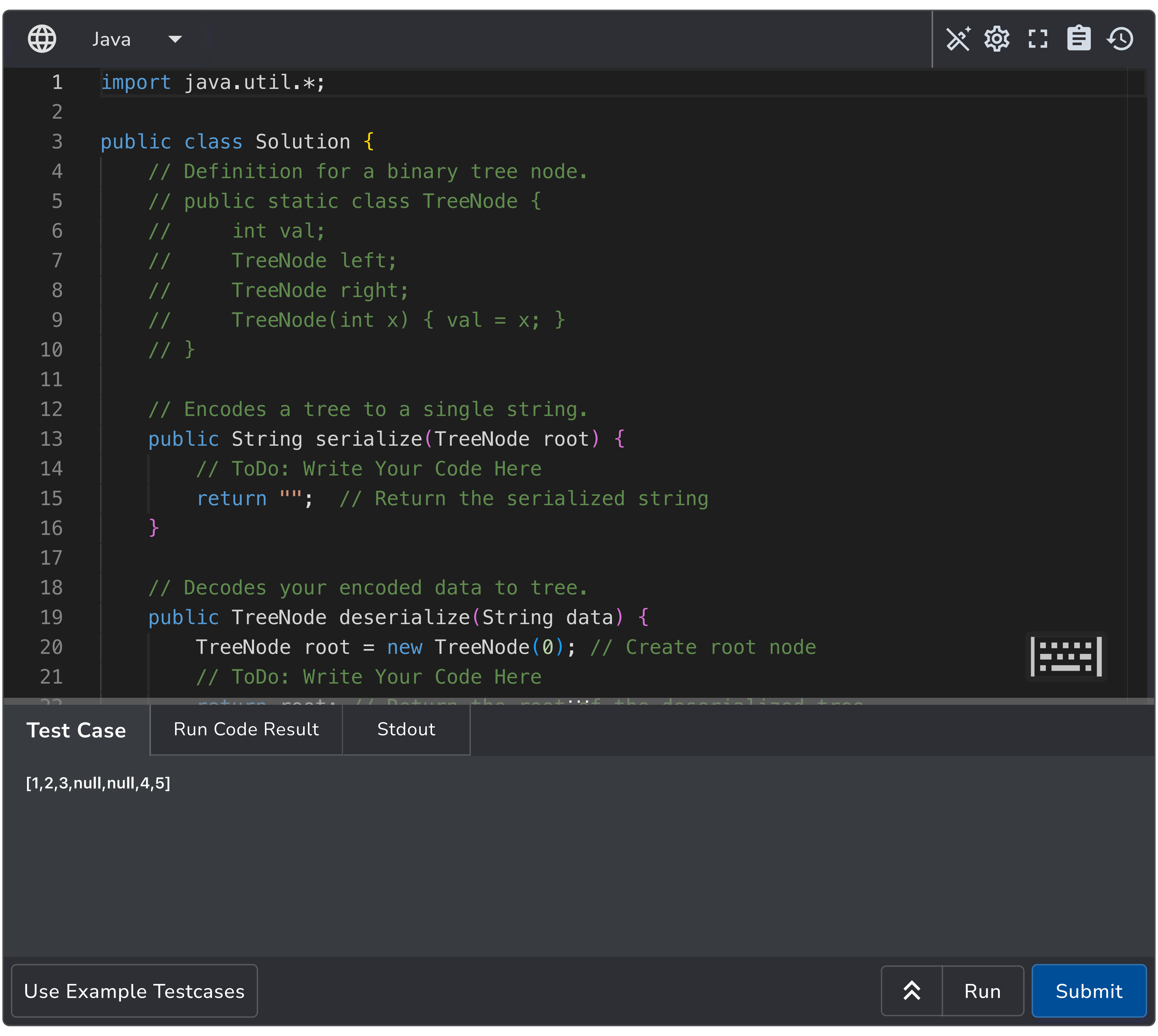
- **Expected Output**: `[5,4,null,3,null,2,null,1]`
- **Justification**: This represents a skewed tree where each node except the last one has a left child. The serialized format follows the tree structure exactly.

Example 3:

- **Input**: `[]`
- **Expected Output**: `[]`
- **Justification**: Since the tree is empty, both the serialized and deserialized outputs should be empty.

Try it yourself

Try solving this question here:



```java
import java.util.*;

public class Solution {
    // Definition for a binary tree node.
    // public static class TreeNode {
    //     int val;
    //     TreeNode left;
    //     TreeNode right;
    //     TreeNode(int x) { val = x; }
    // }

    // Encodes a tree to a single string.
    public String serialize(TreeNode root) {
        // ToDo: Write Your Code Here
        return "";  // Return the serialized string
    }

    // Decodes your encoded data to tree.
    public TreeNode deserialize(String data) {
        TreeNode root = new TreeNode(0); // Create root node
        // ToDo: Write Your Code Here
```

```
[1,2,3,null,null,4,5]
```

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Solution: Serialize And Deserialize Binary Tree

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